Independent Analytical and Hardware-Calibrated Model-Form Validation

Status and claim boundary

This is a **retrospective, post hoc validation campaign** performed after the primary manuscript results were available. It was not preregistered. It uses archived policy surfaces and previously collected hardware telemetry; no new hardware run was performed.

The campaign tests implementation reproducibility, analytical approximation limits, policy-ranking stability, service-law transport, low-fidelity model-form sensitivity, and independent numerical triangulation. It does **not** provide end-to-end controller execution, hardware-in-the-loop validation, or production-safety evidence.

Principal findings

1. Independent optimizer reimplementation

Method	Runs within 1%	Runs within 5%	Maximum excess	Exact training-policy recovery
DR-BGS	100.0%	100.0%	0.348%	88.9%
Guarded GP	100.0%	100.0%	0.878%	86.1%

A clean implementation of the fixed anchors, Matérn-5/2 residual GP, noise handling, lower-confidence acquisition, and five initialization seeds reproduced the manuscript's 0.348% and 0.878% worst cases.

2. Policy-surface validity

- Median diffusion-versus-holdout Spearman correlation: **0.989** (scenario-bootstrap 95% interval 0.977–0.992).
- 10th-percentile Spearman correlation: **0.943**.
- Median Kendall correlation: **0.921**.
- Median diffusion/holdout top-20 overlap: **85.0%**.
- Exhaustive training optimum contained in the diffusion top 20: **86.1%** of scenarios.
- Median scenario-level absolute objective error: **33.4%**.

The diffusion-best policy exactly matches exhaustive training selection in only 50.0% of scenarios; its 90th-percentile and maximum holdout excesses are 5.68% and 13.86%. This validates the paper's use of the diffusion as a trend rather than a final selector or risk estimator.

3. Analytical jump-error indicators

For an interior smooth test function, the third-order Taylor remainder of the compound-jump generator satisfies

$$|R_3 f| \le \frac{M_3}{6}\|f'''\|_\infty,$$

where M_2 and M_3 are arrival-rate-weighted second and third raw jump moments. Scaling derivatives by capacity gives the dimensionless indicator $M_3/(3KM_2)$.

- Median third-order indicator: **0.075**.
- Maximum third-order indicator: **0.216**.
- Median protected-jump probability of exceeding the final 5% of capacity: **0.484**.
- Median eligible-jump probability of exceeding the final 5% of capacity: **0.404**.

Jump coefficient of variation has a Spearman association of 0.438 with rank degradation; the third-order indicator has an association of 0.348. These are diagnostic associations, not causal results.

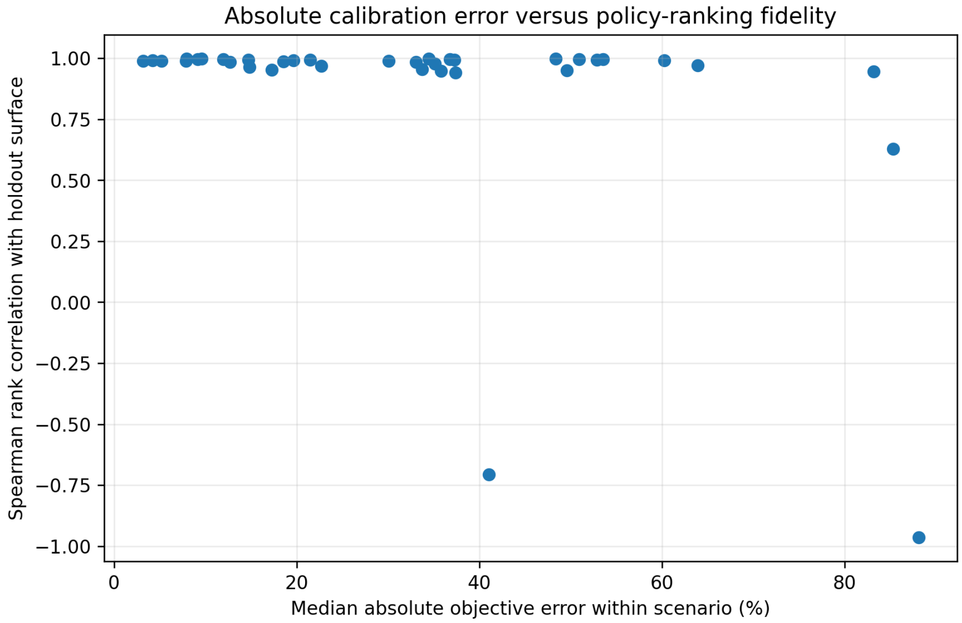

Figure 1: Absolute calibration error versus policy-ranking fidelity

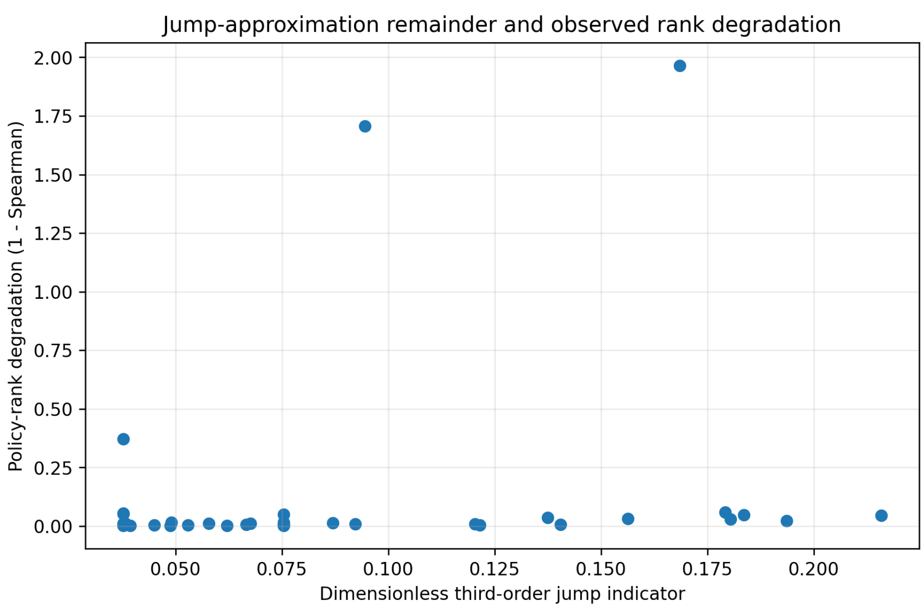

Figure 2: Analytical jump indicator versus policy-rank degradation

4. Direct hardware service-law validation

The Qwen2.5-7B / RTX 3090 direct-telemetry fit is $\widehat{\mu}(z) = 53.6138 \exp(-0.034451z)$.

The exponential model has RMSE 0.789 tokens/s, $R^2 = 0.9961$, AICc 5.16, and leave-one-level-out RMSE 1.299 tokens/s. It outperforms the reciprocal, linear, and quadratic alternatives on AICc or leave-one-out prediction.

A 10,000-draw parametric bootstrap gives intercept interval 53.376–53.857 and decay interval 0.03326–0.03567. A separately aggregated RTX 3090 curve differs from the principal normalized curve by RMSE 0.0099, with no ranking reversal.

5. Cross-hardware and cross-model transport

All three primary domains—Qwen/RTX 3090, Mistral/RTX 3090, and Qwen/RTX 4090—show strictly monotone decline across concurrency.

- Pairwise normalized-curve RMSE: **0.067–0.133**.
- Leave-one-domain-out normalized RMSE: **0.103–0.131**.
- Shared normalized exponential R^2: **0.812**.

The decreasing form transfers more consistently than its coefficient. The evidence supports model-form grounding, not coefficient transfer or controller validation.

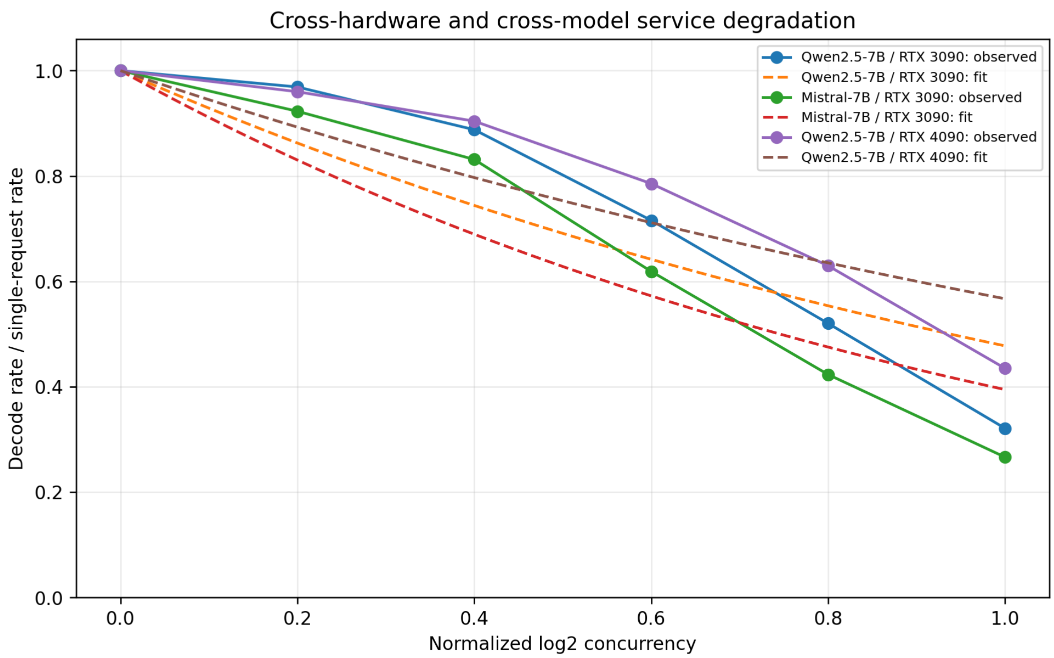

Figure 3: Cross-hardware and cross-model service degradation

6. Low-fidelity model-form stress

Positive affine transformation leaves the selection result unchanged, as expected from affine trend recalibration. Monotone log/rank transformations and smooth coordinate perturbations change some selections, but every tested variant remains within 5% of exhaustive training selection in all 180 runs.

The strongest smooth perturbation has median Kendall correlation 0.944 with the original low-fidelity ranking and maximum holdout excess 4.29%. These are heuristic post hoc stress tests, not a physical replacement for rerunning the simulator under a new service law.

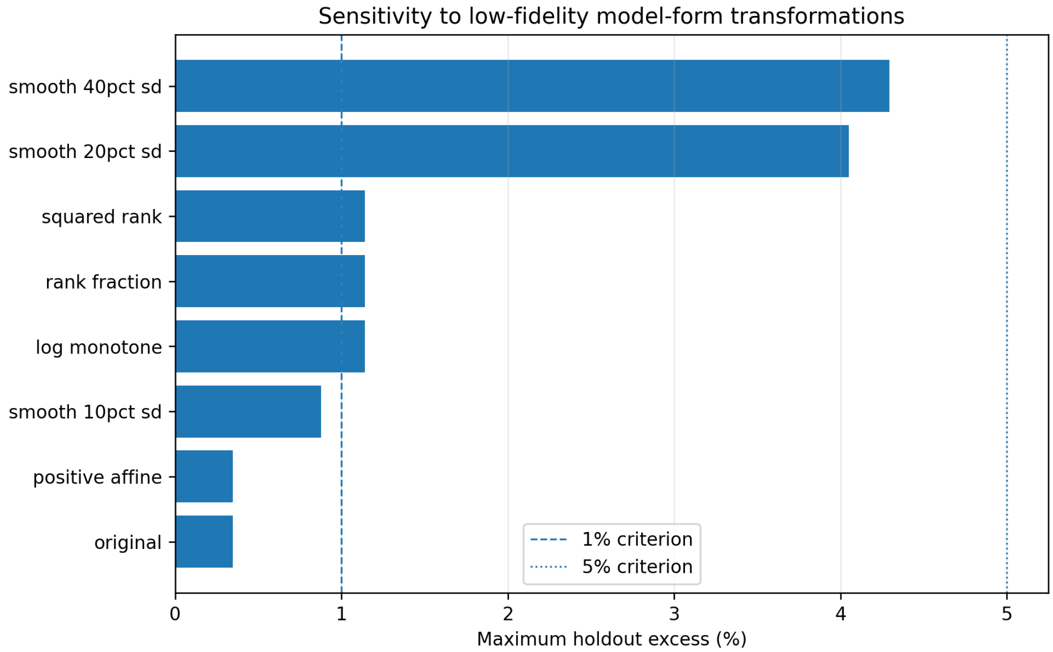

Figure 4: Low-fidelity model-form stress

7. Independent solver triangulation

An independently implemented CTMC solver was applied to 18 unique policy cases across six factorial, robustness, and fresh scenarios. A separate Euler–Maruyama reflected-diffusion simulation supplied a stochastic cross-check.

- Median relative discrepancy between the independent 161-point CTMC and archived objective: **below 0.0001%**.
- Maximum CTMC/archive discrepancy: **0.608%**.
- Median and maximum 161-to-321 grid changes: **1.17%** and **2.25%**.
- Monte Carlo versus 321-point CTMC: median **2.52%**, 90th percentile **4.94%**, maximum **6.67%**.
- Median within-scenario policy-rank Spearman: **1.0**.
- Best-policy agreement: **83.3%**.

The one best-policy disagreement occurs among near-tied policies. The triangulation is consistent at policy scale but is not an exact proof of equivalence.

Overall interpretation

The campaign materially strengthens implementation reproducibility, numerical triangulation, the mathematical explanation of diffusion error, policy-ranking robustness, and cross-hardware/model service-law shape grounding.

It does not remove the central empirical limitation: the complete activation–release controller has not been executed end to end on a live cloud–edge system. The defensible label is **retrospective analytical and hardware-calibrated model-form validation**.

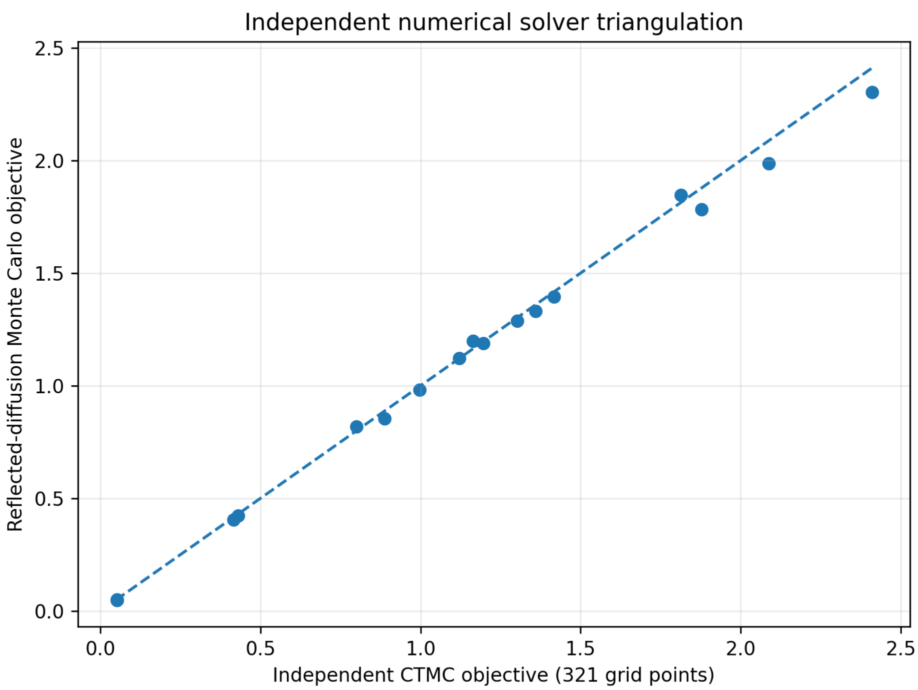

Figure 5: Independent numerical solver triangulation